

The use of grey-based methods in multi-criteria decision analysis for the evaluation of sustainable energy systems: A review



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ABSTRACT

The European energy system is characterized by a high dependence on fossil fuels and energy importation. Renewable energies are a solution to this issue. To change the energy supply, it is necessary to analyze the energy system. Thus, decision-makers face a multi-attribute decision problem. Many sustainable energy systems analyses are characterized by multiple, interrelated parameters and vague information. When the information is vague, it is necessary to implement multi-criteria decision analysis (MCDA) tools that can work with poor data systems. This article reviews appropriated methods and different stages of MCDA problems using grey relation methods (i.e., criteria selection and weighting methods). It is observed that technical criteria are the most evaluated parameters, followed by environmental criteria, in terms of gas emissions. In referring to criteria weighting, the analytical hierarchy process is the most popular tool, especially when it is essential to evaluate the opinion of an expert panel.

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1. Introduction

Changes in the consumption models and the appearance of new extraction technologies (e.g., fracking) have necessitated a series of modifications for taking advantage of energy resources,

the distribution of those energy resources in different territories, and the way that energy is used.

The world's energy map has changed dramatically during the last 40 years (Fig. 1), not only in the increase in the demand for all of the energy sources but also in the contributions of each source at a global level. Thus, at a European level, oil has changed from being the main energy source to sharing the first position with natural gas and coal [1]. The situation in the European area is equivalent to the situation in the whole world. Currently, the EU is undergoing some challenges, including an economic crisis and the growing atmospheric concentration of CO₂, in addition to a

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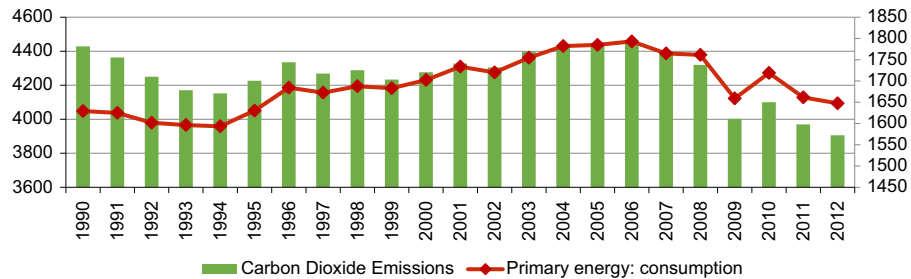


Fig. 1. Impact of the economic crisis on energy consumption (Mtoe) and CO₂ emissions (million tonnes carbon dioxide) in EU. Own preparation based on: BP Statistical Review of World Energy [1].

specific and very preoccupying concern: the insecurity in the energy supply due to a great dependence on outside energy sources [2]. To be able to face this situation, it is necessary to establish some decision criteria that can improve the energy efficiency of all of the processes and activities in fostering the use of renewable energies [3–5].

The European and global energy systems are facing the challenge of choosing the most efficient technology or technologies to be able to address the growing demand. In this respect, despite renewable energy sources are proposed as the solution to European energy problem, represent a small part of total consumption. All of this despite the energy policies have achieved that renewables energy consumption (geothermal, Biomass and other) have increased 71.8% in European Union from 2000 to 2012 [1]. The problem with choosing one technology or another lies with the fact that we are dealing with multi-criteria environments, which are influenced by several interrelated parameters [6]. That is, we need to incorporate environmental, social, technological and economic considerations; thus, we are facing a multi-criteria decision-making (MCDM) problem [7,8]. And to all this must be added the fact, that rational decision-making in energy systems is affected by complex interactions between parameters (i.e. technology, policy and economy). Multi-criteria decision analysis (MCDA) widespread use to analyze energy systems is due to two factors: it provides a method to reduce difficulty for address complex issues with low requirements (i.e. multiple and inter-related objectives, uncertainty, quantifiable or non-quantifiable criteria), and these generally are user-friendly tools (i.e. grey relational analysis) [9]. Literature review also indicates that MCDM methods are indicated to analyze the socio-economic scenario that is undergoing constant changes.

These are the reasons why combining multiple objective optimization and multi criteria decision making analysis play an important role in designing energy systems [10]. In fact, it is well known that the research project design is the most critical and difficult stage, since the success of the project depends on this. In fact, as expected, multi-objective optimization problems always require simultaneously satisfy different and frequently conflicting objectives [11]. Moreover, it must be pointed out that there is no mixture of decision variables values that optimizes simultaneously all factors of the objective vector [10,12].

Although the MCDM problems are highly heterogeneous, their analysis can be structured in three stages (Fig. 2).

Multi-criteria decision analysis methods are used as tools for the classification and selection of the best alternative among a set of options [13,14]. That is, these methods are used to establish the order of the different alternatives once the weights of the selection criteria have been established [9]. As an example, a scheme of the decision hierarchy is shown in Fig. 3. One of the distinctive features of the MCDA method is that it is associated with a limited number of alternatives and preference ratings [15]. That is, MCDAs

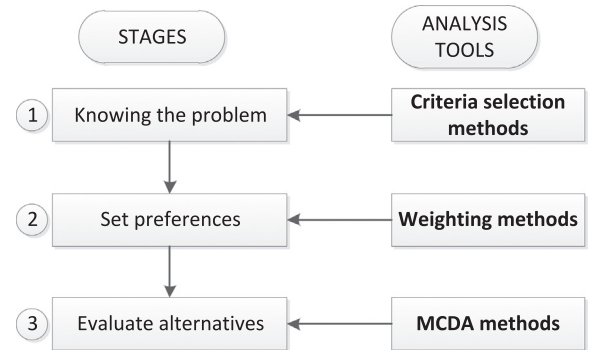


Fig. 2. MCDM problem process.

comprise a set of procedures that allow us to design, assess, and prioritize alternatives in decision problems [16].

One of the problems with analyzing an MCDA system is a lack of data, especially in empirical works. To address this constraint, grey relational methods have been applied in several research works in the energy-environment field [17–32]. This method is used for sustainability assessment of energy systems [20,24], selecting best technology [21,23,27], deciding optimum system option to produce cooling, heating and power [17–19,28,30], planning system management [31], or predict the yield of a system [22]. The application areas have common features of linguistic variables, long-term projects, technical demands, conflicting objectives and uncertainty.

Grey relational analysis (GRA) is a branch of grey system theory [6]. One of the features that distinguishes GRA from conventional statistical methods is that this approach enables assessing quantitative and qualitative relationships between factors and variables using a relatively small amount of data [33]. Although grey system theory was initially focused on the analysis and optimization of uncertainty problems [34–36], it has subsequently been successfully applied in solving forecast problems [37–39]. Therefore, the research directions of grey system theory may be divided into two main lines: relational analysis and prediction. GRA applied to the MCDA process is presented and reviewed in following sections.

1.1. Grey relational analysis: Theoretical basis

The whole concept of grey relational analysis may be summarized as shown in Fig. 4.

As shown on Fig. 4, GRA first step is to normalize the measurements in the range between zero and one. That is, data are normalized once the reference sequence is defined: the highest value is the optimal value or the lowest value is the optimal value. The highest normalized values correspond to the best experimental variable values being equal to the ideal value of 1. The purpose of this measurement

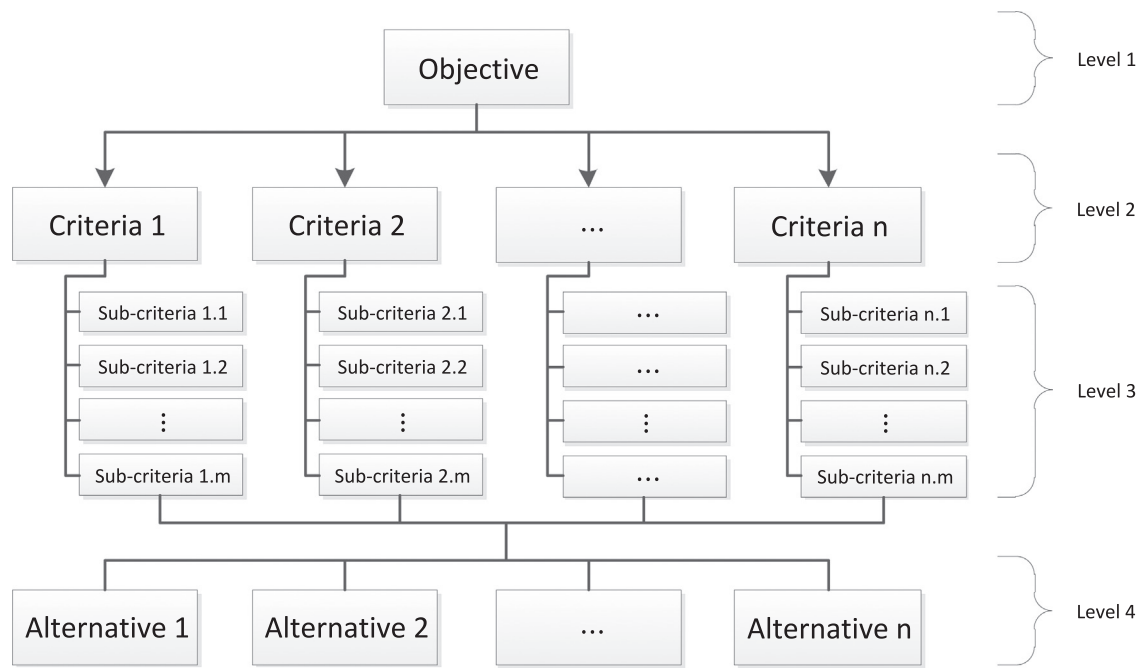


Fig. 3. Decision hierarchy.

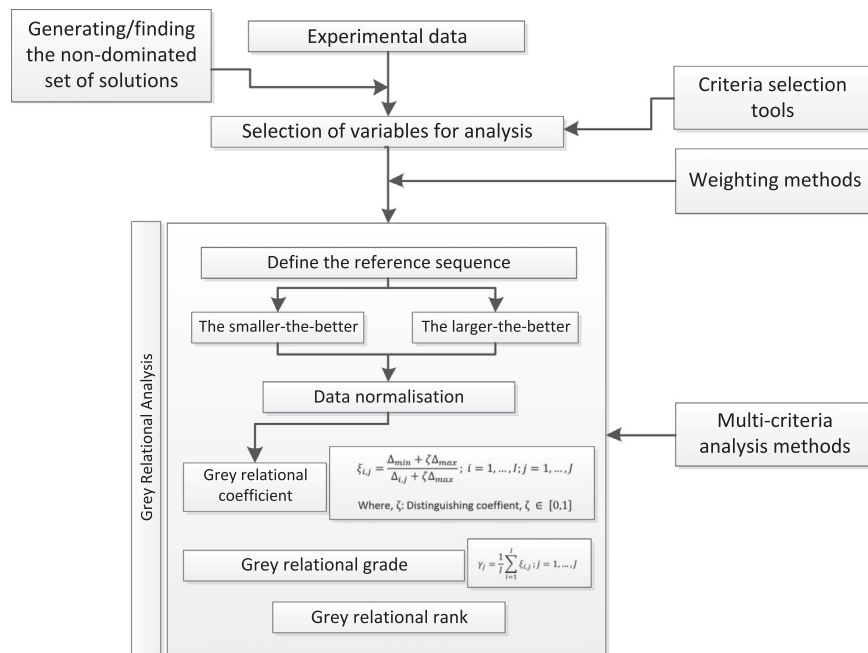


Fig. 4. GRA process.

pre-processing is to have dimensionless variables in order to compare the different sequences.

The best alternative is selected based on the grey relational grade (GRG) value. The GRG indicates the degree of correlation between the reference sequence (at a maximum or minimum point) and a compared sequence [40]. That is, the higher the GRG is, the more similarity exists between the reference sequence and the sequence to be compared. The grey relational grade may also be understood as the degree of influence of the sequence to be compared on the reference sequence [41]. Therefore, grey relational analysis is a measure of the difference between data sequences, which may be used to analyze the degree of correlation between sequences and to filter the data.

2. Grey relational analysis and multi-criteria decision analysis

The recent extensive literature on the application of grey relational analysis to energy issues underlines the current importance of the topic [18,20,24,42–57] because GRA is well suited to the analysis of multi-objective systems [11,58,59]. As other studies have noted, multi-criteria optimization analysis methods have gained in popularity beyond 1990s [9]. Selecting fuels [42,45,48,54,58], optimize operation conditions [44,46,47,53,60], environmental impact assessment, in terms of emissions [49,50,55,61] are the main applications identified in the recent literature. Moreover, it is well recognized that the GRA method in several research fields other than energy to deal with project selection, prediction analysis, performance assessment

or factor effect evaluation [30]. It is common combine GRA method with other techniques such as Analytical Hierarchy Process (AHP) [18], Fuzzy method, Taguchi method [46], Entropy measurement method [43,57], or the combination of different techniques [42].

2.1. GRA and criteria

According to the MCDM problem process, the first step is to know the problem; that is, the first step is to select criteria (see Fig. 2). Based on the criterion classification of Wang et al. [6], the GRA studies in the energy field may be grouped into four criteria: technical, economic, environmental and social (Table 1).

Technical criteria

- **Efficiency:** This approach is expressed in terms of the performance to address the increasing energy demand. This index is the most widely used to evaluate energetic systems [19].
- **Exergy efficiency:** This criterion relates the real thermal efficiency to the highest reversible thermal efficiency. Exergy efficiency may be understood in terms of work. Although exergetic analysis was initially exclusively implemented in the field of industrial ecology, in recent years it has been used in other fields, such as thermal engines [60].
- **Primary energy ratio:** This term is defined as the relationship between the usable energy and the primary energy. Papers related to this issue have focused their research on calculating the energy savings of new energy technologies.
- **Maturity:** The studies included in this section analyze the relative technology maturity, technology properties, regulation and control of optimum technology parameters, system reliability, and other factors.

Economic criteria

- **Investment costs:** Initial costs or investment costs range from equipment and plant construction costs to the variation in the capacity of a specific port facility.
- **Operational and maintenance costs:** The operational and maintenance costs include both staff and service costs. The analytical study of the maintenance costs aims to increase the product life cycle and prevent potential failures.

Table 1

Research classification based on performance measures and performance criteria.

Criteria	Measures	Literatures
Technical	Efficiency	[20,42,45,53,62–74]
	Exergy efficiency	[60,64]
	Primary energy ratio	[40,69]
	Maturity	[45–48,63,64,67,74–79]
Economic	Investment costs	[64,72,80,81]
	Operational and maintenance costs	[21,24,70,72,82]
	Fuel costs	[65,83]
Environmental	Gas emissions	[20,21,60,64,65]
Social	Human/technological impact	[24,61,84]

Table 2

Research classification based on analysis method.

Methods	Literature
GRA	[45,47,48,53,60–63,67–69,73–76,80,81,83]
GRA combined with other tools	AHP [20,70,71,82] Fuzzy method [21,24,40,64,65,84] Taguchi method [46,66] Mix different tools (i.e. TOPSIS, Fuzzy) [42,72,78,79]

- **Fuel costs:** Fuel is the essential and determinative element of any energy system [20,30]. It is therefore appropriate to assess its importance, especially considering the dramatic fluctuations in fuel prices.

Environmental criteria

- **Gas emissions:** This index assesses the level of pollution caused by the energy system on the environment in terms of emissions (NO_x, CO and CO₂).
- **Human/technological impact:** Human impact refers to the negative impact on the ecosystem by the energy consumption of human production and/or the positive impact on the ecosystem by human environmental protection [61]. The technical impact may be defined as the negative impact of energy technologies.
- **Social criterion**
- **Social acceptance:** Social acceptance reflects the opinions of experts and/or users in relation to the execution of a particular project.

In general, these parameters are interrelated and work in opposition, which hinders the multi-objective selection process.

It can be observed from the analyzed literature that GRA is widely used combined with other tools (Table 2). The purpose of combine different tools is to fill the gap in GRA methodology. The principal limitation of GRA is that grey numbers are not sufficient to represent complete information (i.e. to give linguistic assessment), as they are represented by a range, instead of crisp value [21,27,72].

The previous step to selecting the criteria is to choose the study variables. For this phase, support vector machines (SVMs) are becoming increasingly important combined with the subsequent grey relational analysis [85].

2.2. Combined tools: GRA and weighting methods

Grey relational analysis is a mathematical tool to select the best alternative in multi-objective systems (step 3 in Fig. 2). However, GRA fails to distinguish the relative importance of the different sequences. Therefore, GRA has been combined with objective and subjective weighting methods. Objective weighting methods are based on data variables (information that weakens subjectivity), whereas subjective weighting methods attribute weights based on expert preferences for attributes [86].

2.2.1. GRA and objective weighting methods

2.2.1.1. GRA and TOPSIS. The technique for order of preference by similarity to ideal solution (TOPSIS) can be used both as a weighting method and as an MCDA method [6,87,88]. When TOPSIS is used as an MCDA method, the best solution is one that one is closest (by Euclidean distance) to the positive ideal solution and farthest from the negative ideal solution [15,89]. Thus, the TOPSIS methodology assumes that each criterion has a (positive or negative) regular variation of its utility [87,90]. When GRA and TOPSIS are used together, TOPSIS acts as a criteria-weighting tool [42,72].

Eq. (1) shows a TOPSIS-GRA information matrix. $A = \{A_i | i = 1, \dots, n\}$ represents the set of alternatives; $C = \{C_j | j = 1, \dots, n\}$, the set of criteria; $X = \{x_{ij} | i = 1, \dots, n; j = 1, \dots, n\}$, the set of performance ratings and $W = \{W_j | j = 1, \dots, n\}$, the set of weights.

Alternatives	C_1	C_2	$\dots$	C_n	
A_1	x_{11}	x_{12}	$\dots$	x_{1n}	
A_2	x_{21}	x_{22}	$\dots$	x_{2n}	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
A_n	x_{n1}	x_{n2}	$\dots$	x_{nn}	
W	w_1	w_2	$\dots$	w_m	(1)

Criteria weights are assigned by the following equation (using the TOPSIS method) [91,92]:

$$w = \left(\frac{1}{\sum_{i=1}^m \frac{1}{s_i}}, \frac{1}{\sum_{i=1}^m \frac{1}{s_2}}, \dots, \frac{1}{\sum_{i=1}^m \frac{1}{s_m}} \right)^T \quad (2)$$

where s_i is the square of the weighted distance between the n alternatives and the ideal alternative.

In energy system analysis, is common to have both objective and subjective criteria in the decision matrix (Eq. (1)), and it is usual to combine objective and subjective weighting tools. One example is the work of Zheng et al. [72]. In this research study, the building-envelopment variables for saving energy are analyzed. To analyze these variables, the analysis factors are grouped around six criteria: thermal performance, building form, economy, innovation, reliability and environmental protection. Without going deeper into these parameters, it can be seen that some of them are qualitative indices. Therefore, the TOPSIS tool is unsuited to analyze this problem. The solution proposed by Zheng et al. [72] consists of combining TOPSIS and the matrix analysis method (based on the analytical hierarchy process (AHP), which will be discussed in later sections). This methodology was also used by Chen and Hao to analyze energy markets [93]. Both research studies used the work of Wei and Feng as a Ref. [94].

2.2.1.2. GRA and fuzzy methodology. Fuzzy methodology may be used with either objective or subjective weighting methods. Therefore, Jing et al. [64], in an optimization-combined cooling, heating and power (CCHP) study, combined entropy methods with fuzzy methods. Because the existing literature in the energy systems field uses this method to address subjective criteria, the fuzzy method was included in this section.

GRA combined with fuzzy methodology is applied to evaluate qualitative parameters by expert-panel assessments [24,42,65, 84,95–101]. Fuzzy methodology is used to associate a numerical value with a linguistic criterion. The main advantage of fuzzy methodology compared with the AHP is that it allows incorporating incomplete or vague information into the decision-making process [102]. This method works very similarly to the AHP. The major difference between them concerns the triangular vectors composed of triangular fuzzy numbers (TFN). The rating scale (Saaty's scale) consists of nine levels where the relative importance of each criterion is assigned based on expert assessment (i.e.,

"equal importance" is associated with the triangular fuzzy vector (1,1,1)). The scale ranges from equal importance to extreme importance. In a paper about hydrogen energy storage (HES) techniques, Gumus et al. [103] apply AHPs to improve fuzzy-GRA behavior. In this paper, alternatives were selected based on based on expert opinions and a literature review, which often involves subjective data and/or vague information (i.e., the difficulty of providing deterministic preferences). The AHP is applied to weighting criteria and fuzzy methods to include different types of information. The technique is known as the fuzzy extension of the AHP (FAHP). Manzardo et al. [21], to select the best renewable energy technology through a life cycle sustainability perspective, modified the GRA method to include numerical intervals (i.e., linguistic assessments) in evaluating the alternatives. This approach is particularly useful when assessing energy technologies, where decision-makers evaluate the alternatives using linguistic terms (e.g., "bad" or "good").

2.2.1.3. GRA and entropy method. The entropy method is the most representative objective weighting tool. This method determines the relative importance of the attributes and how they are related to the evaluation or final result [57,104]. That is, the entropy shows how a criterion reflects the information system and its uncertainty [30,105]. The weight of the normalized criteria is calculated from the following equation [46,106]:

$$w_i = \frac{1 - e(x_i)}{\sum_{i=1}^m (1 - e(x_i))} \quad (3)$$

where $e(x_i) \geq 0$ represents the i th criterion entropy $y \sum_{i=1}^m w_i = 1$

Entropy is calculated based on the following equation:

$$e(x_i) = - \frac{\sum_{k=1}^m f_{ki} \ln f_{ki}}{\ln m} \quad i = 1, \dots, n \quad (4)$$

where f_{ki} is traditionally calculated based on:

$$f_{ki} = \frac{b_{ki}}{\sum_{i=1}^m b_{ki}} \quad (5)$$

Using b_{ki} in the process of standardizing raw data:

$$b_{ki} = \frac{\max_k x_{ki} - x_{ki}}{\max_k x_{ki} - \min_k x_{ki}} \quad (6)$$

This tool is useful in studies evaluating several criteria according to easily measurable and verifiable objective parameters [72]; in other words, it is used in papers focused on variables such as gas emissions, energy consumption, and failures, among others. It is not recommended to use the entropy method in economic analyses for which the variables fluctuate or where not all of the information is known (e.g., in cases of professional secrecy). For instance, Hua et al. [57], in a study about power system faults, used the entropy method to build a weight matrix to analyze the parameters that may cause system failure (e.g., the duration of low voltage, the voltage deviation, the quantities of active and reactive power shedding, etc.).

2.2.2. GRA and subjective weighting methods

2.2.2.1. GRA and pair-wise comparison. Ilankumaran et al. [95], in a paper about wastewater treatment, used the pair-wise comparison methodology to determine relative weights for criteria and sub-criteria (Fig. 3, levels 2 and 3). The pair-wise comparison methodology is based on an expert panel's comparison of the relative importance of two criteria [72,107,108]. Although several different scales are available, Saaty's scale is most often used in the pair-wise method [109] to build weight matrices. This scale is built upon nine levels of

Table 3
Saaty's scale in pair-wise comparison [110].

Importance intensity	Relative importance
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong or demonstrated importance
9	Extreme importance
2,4,6,8	Intermediate values

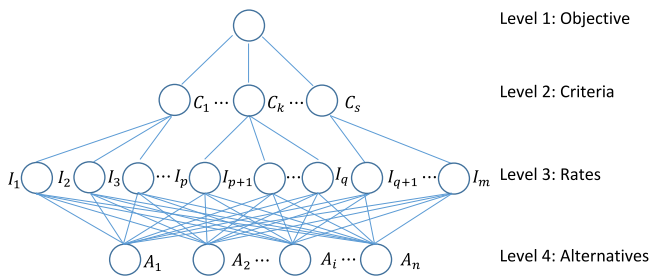


Fig. 5. AHP hierarchical system.

importance intensity, from equally important to absolutely more important (Table 3). For example, if the panel of experts selects the intensity level of 1 (i.e., equally important) the description would mean that the two compared criteria contribute equally to the final goal. Within the energy sector, this tool is generally employed in papers that analyze environmental indices and the emissions sub-criterion in particular [20,23]. This tool is used because of the existence of penalties and internal market rules, which may be taken into account only in pair-wise comparisons.

2.2.2.2. GRA and AHP. The analytical hierarchy process (AHP) is the subjective weighting method most widely used for analyzing energy systems [18–20,23,24,26,70–72,88,103,111–116]. The main advantage of this method, explaining its widespread use, is that it handles multiple-criteria problems easily. The AHP method decomposes a multi-criteria decision problem into a hierarchy [117]. The AHP methodology starts off with an expert selection process and evaluation criteria. Therefore, this technique is based on the pair-wise comparison model. A hierarchical model can be defined as a tree, where the main objective is on the treetop and the alternatives are the roots (Fig. 5).

In the hybrid method (AHP–GRA), the AHP is used to obtain the relative weights, and GRA is used to rank the overall system performance (Fig. 6). This combined method comprises the following steps [70]:

1. The AHP is used to build the selection hierarchy matrix and establish the weights of the alternatives [84,86]. Once the matrix is built, the criteria weights are calculated by three main methods: the arithmetic mean, the characteristic root and the least-squares method [118].
2. The grey relational coefficients are determined through GRA. The grey relational grades are calculated as the sum of the products of the grey relational coefficients and their corresponding weights.

The AHP is commonly used to evaluate energy sustainability based on economic and environmental criteria [19,20,24,113] because the AHP is very useful when different energy systems are analyzed and there are conflicting goals. In addition to being an easy-to-understand technique, the AHP allows perception judgment intervals to be included [119]. This inclusion is particularly important in energy analyses that provide deterministic preferences for decision makers, which may sometimes be difficult (Fig. 7). For instance, this method can be used to choose which gas emission is most important: CO₂, NO_x or CO [19].

2.2.3. GRA and Delphi method

The Delphi method may be used to select criteria such as assigning weights [120–130]. This technique is based on obtaining a reliable expert-panel consensus view on a specific topic [131,132]. For this technique, questionnaires are given in several

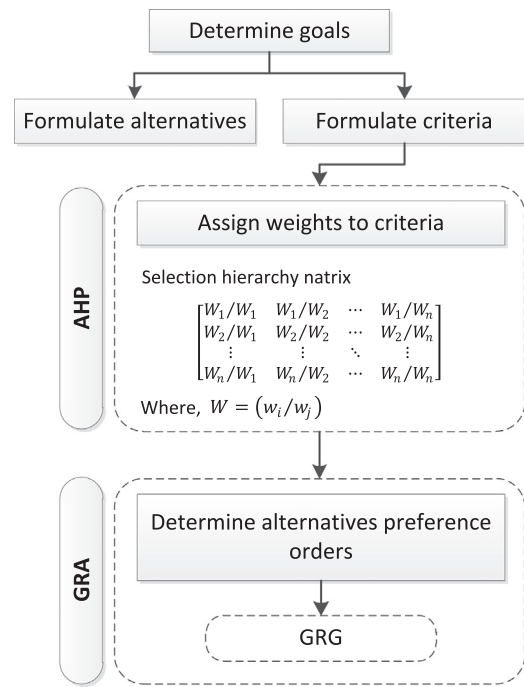


Fig. 6. Hybrid method: AHP and GRA.

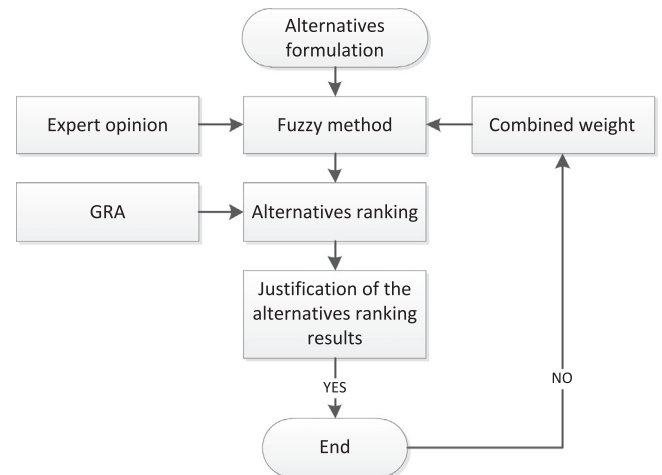


Fig. 7. GRA and fuzzy methodology scheme [62].

rounds. Experts are individually interviewed at the end of each round to justify their answers. This information feedback is used in the next round of questions [133]. Therefore, the expert answers are more reliable. The Delphi method, as a weighting tool to subsequent GRA selection, may be used by itself [134–136] or combined with subjective weighting tools as fuzzy methods and/or AHPs [21,122,137]. The strength of this method is that linguistic terms may be included in the analysis energy systems. In addition, the Delphi method is a useful tool when there are scheduling problems or when it is not possible to group together all of the experts (i.e., experts in hydrogen production technologies in different countries) [122].

2.2.4. GRA and other subjective weighting methods

Zhao et al. [71] proposed an alternative method to evaluate subjective elements: a binary domain matrix. Each criterion weight is calculated based on an indicator aggregate score divided by the total

indicator score:

$$W = \frac{x_i + 1}{\sum_{i=1}^n x_i + 1} \quad (7)$$

where W is the weight, and x_i is the score for each criterion.

Riva and Silva-Rocha [113] applied ' method combined with AHPs and the pair-wise comparison method to classify Brazilian favelas for an energy efficiency program. This tool was proposed by Simos [138], and its innovation consists of associating a "playing card" with each criteria. That is, the expert panel classifies the cards (i.e., the criteria) from least important to most important. With the aim to include the relative importance (i.e., preferences between criteria), coloured cards are added.

The eigenvector method may be applied to evaluate relative weights; however, it is not a widely used technique. To analyze wastewater techniques, Zeng, Jiang, Xu and Li [95] used the eigenvalue method combined with AHPs to compute the weight of each criterion once the pairwise comparison matrix was established based on economic, technical and administrative criteria.

3. Conclusion

In this article, we have reviewed the studies linked to the energy field that analyze multi-criterion environments under the condition of uncertain information through the grey relation method, reaching the following conclusions:

1. In the analysis of multi-objective systems, the most-used criterion is the technical criterion, and more specifically, the efficiency of the energy system is the most-used sub-criterion. The environmental criterion, in terms of gas emissions (CO_2 , NO_x and CO), is the criterion that has been analyzed second-most.
2. In most of the investigations of energy systems in which GRA is used, it is used with tools to evaluate weights with linguistic preferences. That is, GRA is used with studies in which part of the information is provided by a panel of experts.
3. A trend to modify the initial GRA models can be observed in recent studies, combining them with other tools or adding improvements to them. Regarding this trend, it can be observed that there are already several projects in which the tool has been modified [21,58,139–141]
4. Despite being a recently created analysis tool, GRA has had wide expansion and deployment. We note that the policies for supporting the energy sector have contributed to that success as a large number of the studies have been conducted in the last 10 years.

Therefore, we can state that GRA is a powerful tool for decision-making in energy systems characterized by the shortage of information. Moreover, validation of results with multiple methods to tackle data uncertainty increase applicability and powerful of GRA [6,9].

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